

Particle-resolved LBM-DEM simulations of sheared suspensions using Lees–Edwards boundary conditions

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We present a particle-resolved simulation method for analyzing the rheology of suspensions consisting of noncolloidal, monodisperse, spherical particles dispersed in an incompressible Newtonian fluid. The method employs the lattice Boltzmann method for fluid flow and the discrete element method for particle dynamics, coupled through the improved smoothed profile approach. Lees–Edwards (LE) boundary conditions are implemented to enable efficient simulations of suspensions under shear flow. Verification tests confirm the correct implementations of the LE boundary conditions and the method for evaluating the apparent viscosity of suspensions. Applications to suspension shear flows over a wide range of particle volume fractions and interparticle friction coefficients show good agreement with previous experimental and numerical results, and capture the transition in the dominant mechanisms controlling suspension viscosity with increasing particle volume fraction, from the suspending fluid viscosity, to fluid–particle hydrodynamic interactions, and finally to particle–particle direct contacts. These results demonstrate that the proposed method provides an efficient, robust, and accurate tool for numerical investigations of suspension rheology.

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I. INTRODUCTION

Suspensions of solid particles in fluids are found in a wide range of natural and industrial systems. Predicting their flow behavior is crucial for effectively designing and controlling such systems. However, these suspensions exhibit complex rheological behaviors influenced by various factors, including the physical [1–3] and physicochemical [4,5] properties of particle surfaces, the characteristics of the suspending fluid [6], and the particle volume fraction. Even in the simplified case of noncolloidal, monodisperse, spherical particles suspended in a Newtonian fluid, the rheology of the suspension is not fully understood, although considerable research has been devoted to identifying the underlying mechanisms [7–10]. In the present study, we focus on the simplified suspension, in which only hydrodynamic interactions and interparticle physical contacts are

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considered, while other multiphysics effects such as Brownian motion, physicochemical interactions, and non-Newtonian rheology of the suspending fluid are neglected.

Numerical simulation is a powerful tool for investigating the dynamics of particle suspensions at the particle scale. The Stokesian dynamics (SD) method [3,11] and the discrete element method (DEM) [12], which is often combined with simplified SD approaches [1,5,13], are widely used to study suspension rheology. These methods can efficiently simulate flows that involve many particles through elegant simplifications and have successfully captured the fundamental mechanisms underlying these phenomena.

Another promising approach is particle-resolved computational fluid dynamics simulation [14–19]. In this approach, the motion of the suspending fluid is resolved on a grid sufficiently finer than the particle size, whereas particle motion is computed using the molecular dynamics (MD) method or DEM, which are coupled in a two-way manner. This approach enables a more detailed analysis of suspension flows with fewer assumptions than the aforementioned methods. Gallier *et al.* [20] simulated the shear flow of noncolloidal suspensions using a fictitious domain approach, in which the Stokes flow is solved at the particle scale [14], and suggested that particle-resolved simulations might be suitable when detailed flow information (e.g., particle diffusion or velocity correlations) is needed.

Despite their advantages, particle-resolved simulations are computationally expensive, which limits the size of the systems that can be analyzed. Therefore, it is crucial to develop an efficient rheological simulation method for suspensions based on particle-resolved approaches. We recently proposed a particle-resolved simulation method that combines the smoothed profile (SP) method [21,22] with the lattice Boltzmann method (LBM) [23,24], which we referred to as the improved SP-LBM (iSP-LBM) [25]. In this method, we compute hydrodynamic forces on moving solid particles exactly by eliminating the effects of the internal fluid mass within the particle region, which enables the stable and accurate treatment of particle–fluid boundaries in multiparticle systems [25–27]. We expect the iSP-LBM to provide an efficient method for rheological simulations of suspensions.

In rheological simulations of suspensions, shear flow is typically imposed by placing two walls moving in opposite directions, which requires a large computational domain to minimize wall effects [14]. One approach used to reduce the computational cost is the Lees–Edwards (LE) boundary conditions [28], which impose shear flow without physical walls and allow for a smaller computational domain. This technique, first applied to LBM simulations of fluid-only systems by Wagner and Pagonabarraga [29], has been extended to particle suspension flows in two-dimensional (2D) [30–32] and three-dimensional (3D) [16] systems by only a few groups. In this study, we implement the LE boundary conditions in our 3D iSP-LBM method.

This study aims to present a particle-resolved simulation method for analyzing the shear flow of suspensions of noncolloidal, monodisperse, spherical particles in an incompressible Newtonian fluid. Building on our previous work [33], where its basic feasibility was demonstrated under restricted conditions without quantitative verification, the present study establishes a systematically verified and quantitatively reliable method. Key numerical parameters, including particle resolution and the interpolation order for the LE boundary treatment, are determined through verification tests. The method’s numerical stability is also quantitatively evaluated under challenging particle density conditions. The particle–particle contact model is improved by accounting for surface roughness [14]. The method is then applied to a systematic analysis of suspension shear flows and compared with previous experimental and numerical results, highlighting the potential of the proposed method as a tool for quantitative investigations of suspension rheology.

II. NUMERICAL METHOD

We present a particle-resolved simulation method for analyzing the shear flow of suspensions consisting of noncolloidal, monodisperse, spherical particles dispersed in an incompressible Newtonian fluid (Fig. 1). The LBM for fluid dynamics is coupled with the DEM for particle dynamics using

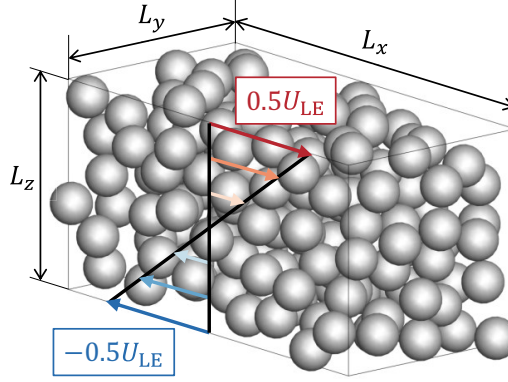


FIG. 1. Computational domain with dimensions $L_x \times L_y \times L_z$ used for simulations of shear flows of particle suspensions under the Lees–Edwards boundary conditions. The imposed shear rate is $\dot{\gamma} = U_{LE}/L_z$, where $\pm U_{LE}$ are the velocities of the upper and lower replica domains moving in the x -direction, respectively.

the iSP method [25]. The iSP-LBM was thoroughly validated in our previous studies, including flow past a circular cylinder, particle sedimentation and flotation, and capillary interactions between particles at a two-fluid interface [25–27]. The LE boundary conditions are implemented in the iSP-LBM-DEM to enable efficient simulation of shear flows. Here, we briefly summarize the key features of the iSP-LBM-DEM method and detail the implementation of the LE boundary conditions.

A. Fluid dynamics

The motion of the suspending fluid is governed by the continuity equation and the Navier–Stokes equations with a forcing term:

$$\hat{\nabla} \cdot \hat{\mathbf{u}} = 0, \quad (1)$$

$$\frac{\partial \hat{\mathbf{u}}}{\partial \hat{t}} + (\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}} = -\frac{1}{\hat{\rho}} \hat{\nabla} \hat{p} + \hat{\nu} \hat{\nabla}^2 \hat{\mathbf{u}} + \alpha \hat{\mathbf{f}}^H, \quad (2)$$

where $\hat{\mathbf{u}}$ is the fluid velocity, $\hat{\rho}$ is the fluid density, $\hat{p}$ is the pressure, and $\hat{\nu}$ is the fluid kinematic viscosity. The term $\alpha \hat{\mathbf{f}}^H$ in Eq. (2) represents the body force that enforces the no-slip conditions on rigid particle surfaces in the SP method [22], which are described in Sec. II C. In this paper, dimensional quantities are denoted by a hat (^); dimensionless variables are written without a hat unless otherwise stated.

These calculations are conducted using the LBM [23,24]. In the LBM, the fluid is modeled as a collection of virtual particles with discrete velocities. In this paper, dimensionless variables are used and defined based on a characteristic length $\hat{L}_0$, characteristic virtual-particle speed $\hat{c}$, characteristic timescale $\hat{t}_0 = \hat{L}_0/\hat{U}_0$, where $\hat{U}_0$ is a characteristic flow speed, and reference fluid density $\hat{\rho}$ [24]. Here, $\hat{c}$ is determined by the lattice spacing and time step, as described later, and is therefore independent of the macroscopic flow speed $\hat{U}_0$.

In this study, a 3D lattice with 15 discrete velocity vectors, known as the D3Q15 model (Fig. 2), is used for computational efficiency. The computational domain is divided into a uniform cubic lattice. The discrete velocity vectors $\mathbf{c}_i$ of the virtual particles are defined as $\mathbf{c}_i = (0, 0, 0), (\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1), (\pm 1, \pm 1, \pm 1)$ for $i = 1, 2, \dots, 15$. All simulations were conducted at very low Reynolds numbers, under which no noticeable artifacts associated with lattice anisotropy were observed, indicating that the D3Q15 model was sufficient for these conditions. At higher Reynolds numbers, higher-order lattice models such as D3Q19 or D3Q27, which can reduce lattice anisotropy, may become advantageous. For the collision model, the single-relaxation-time (SRT)

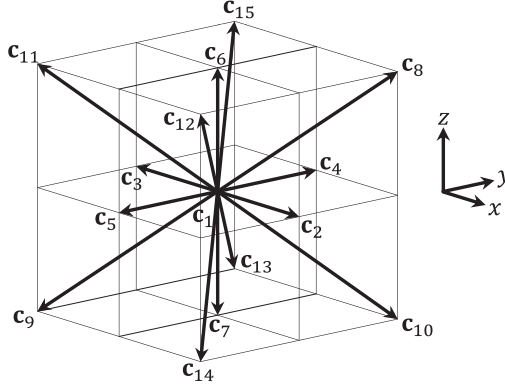


FIG. 2. Three-dimensional 15-velocity (D3Q15) model.

model is adopted for its simplicity and computational efficiency. Under the low-Reynolds-number conditions considered here, the SRT model provides sufficient numerical stability. For higher-Reynolds-number conditions, multiple-relaxation-time (MRT) or two-relaxation-time (TRT) models could be used to further enhance numerical stability.

Based on the SRT model, the time evolution equation for the particle distribution function $f_i(\mathbf{x}, t)$ corresponding to the velocity $\mathbf{c}_i$ at position $\mathbf{x}$ and time t is given by

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta x, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)] + 3\Delta x E_i \mathbf{c}_i \cdot \boldsymbol{\alpha} \mathbf{f}^{\text{H}}(\mathbf{x}, t), \quad (3)$$

where $f_i^{\text{eq}}(\mathbf{x}, t)$ is the local equilibrium distribution function, τ is the relaxation time, Δx is the lattice spacing, and Δt is the time step during which virtual particles travel across one lattice spacing. Note that $\Delta t = \text{Sh} \Delta x$, where the Strouhal number Sh is defined as $\text{Sh} = \hat{L}_0 / (\hat{t}_0 \hat{c}) = \hat{U}_0 / \hat{c}$ [24].

The local equilibrium distribution function $f_i^{\text{eq}}(\mathbf{x}, t)$ for incompressible fluids [24] is given by

$$f_i^{\text{eq}}(\mathbf{x}, t) = E_i \left[3p + 3\mathbf{c}_i \cdot \mathbf{u} + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2}(\mathbf{u} \cdot \mathbf{u}) \right], \quad (4)$$

where $E_1 = 2/9$, $E_2 = \dots = E_7 = 1/9$, $E_8 = \dots = E_{15} = 1/72$. The pressure p and fluid velocity $\mathbf{u}$ are computed from the distribution functions as

$$p(\mathbf{x}, t) = \frac{1}{3} \sum_{i=1}^{15} f_i(\mathbf{x}, t), \quad (5)$$

$$\mathbf{u}(\mathbf{x}, t) = \sum_{i=1}^{15} \mathbf{c}_i f_i(\mathbf{x}, t). \quad (6)$$

The kinematic viscosity ν of the fluid is related to the relaxation time τ by

$$\nu = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \Delta x. \quad (7)$$

B. Particle dynamics

The motion of N spherical particles, with radius a , mass M_p , and moment of inertia I_p , is computed using the DEM [12]. The translational and rotational motions of the k -th particle

($k = 1, 2, \dots, N$) are governed by the Newton–Euler equations:

$$\text{Sh} \frac{d\mathbf{X}_k}{dt} = \mathbf{V}_k, \quad (8)$$

$$M_p \text{Sh} \frac{d\mathbf{V}_k}{dt} = \mathbf{F}_k^H + \mathbf{F}_k^C, \quad (9)$$

$$I_p \text{Sh} \frac{d\boldsymbol{\Omega}_k}{dt} = \mathbf{T}_k^H + \mathbf{T}_k^C, \quad (10)$$

where $\mathbf{X}_k$, $\mathbf{V}_k$, and $\boldsymbol{\Omega}_k$ denote the center-of-mass position, translational velocity, and angular velocity of the k -th particle, respectively. The terms $\mathbf{F}_k^H$ and $\mathbf{T}_k^H$ represent the hydrodynamic force and torque, respectively, which are described in Sec. II C. The terms $\mathbf{F}_k^C$ and $\mathbf{T}_k^C$ represent the contact force and torque, respectively. In this study, the focus is on noncolloidal particles; therefore, physicochemical interparticle forces, such as van der Waals and electrostatic interactions, are neglected. However, such forces can be readily incorporated into Eq. (9), if necessary. The translational and angular velocities of the particles [Eqs. (9) and (10)] are calculated using the first-order explicit Euler scheme, while the particle positions [Eq. (8)] are updated using the Crank–Nicolson scheme.

The contact force $\mathbf{F}_k^C$ and torque $\mathbf{T}_k^C$ acting on the k -th particle are given by

$$\mathbf{F}_k^C = \sum_{\delta_{kl} \geq 0} \mathbf{F}_{kl}^C = \sum_{\delta_{kl} \geq 0} (\mathbf{F}_{kl}^{C,n} + \mathbf{F}_{kl}^{C,t}), \quad (11)$$

$$\mathbf{T}_k^C = \sum_{\delta_{kl} \geq 0} \mathbf{T}_{kl}^C = \sum_{\delta_{kl} \geq 0} a \mathbf{n}_{kl} \times \mathbf{F}_{kl}^{C,t}, \quad (12)$$

where $\mathbf{F}_{kl}^C$ and $\mathbf{T}_{kl}^C$ denote the contact force and torque, respectively, exerted on the k -th particle by the l -th particle ($l = 1, 2, \dots, N$ and $l \neq k$); $\mathbf{F}_{kl}^{C,n}$ and $\mathbf{F}_{kl}^{C,t}$ represent the normal and tangential components of the contact force, respectively; δ_{kl} is the overlap distance between the k -th and l -th particles; and $\mathbf{n}_{kl}$ is the unit normal vector pointing from the center of the k -th particle to the center of the l -th particle. The contact forces are calculated using a linear spring–dashpot model [34,35]. The tangential force obeys Coulomb’s friction law, $|\mathbf{F}_{kl}^{C,t}| \leq \mu_c |\mathbf{F}_{kl}^{C,n}|$, where μ_c is the interparticle coefficient of friction.

In this study, rough particles characterized by a surface roughness height h_r are considered. Contact between the k -th and l -th particles is assumed to occur when the center-to-center distance $|\mathbf{X}_k - \mathbf{X}_l|$ is less than $2a + h_r$ [14]. The overlap distance δ_{kl} is defined as $\delta_{kl} = 2a + h_r - |\mathbf{X}_k - \mathbf{X}_l|$ and is used to calculate the contact forces.

C. Fluid–particle coupling

The LBM and DEM are coupled using the SP approach [21,22,25]. In this approach, the sharp interfaces between the solid particles and the surrounding fluid are replaced by diffuse interfaces of finite thickness. The presence of particles is represented as a continuous scalar field over the computational domain:

$$\alpha(\mathbf{x}, t) = \sum_{k=1}^N \alpha_k(\mathbf{x}, t), \quad (13)$$

where $\alpha_k(\mathbf{x}, t)$ is the SP function for the k -th particle at position $\mathbf{x}$ and time t . This function smoothly transitions from $\alpha_k = 1$ inside the particle to $\alpha_k = 0$ outside the particle, across the diffuse interface of thickness ξ [25].

Assuming that the interior and interfacial regions of particles are filled with an artificial fluid that has the same density and viscosity as the surrounding fluid, the hydrodynamic field is computed over the entire domain using the LBM. The particle velocity field $\mathbf{u}_p(\mathbf{x}, t)$ is defined using the particle

field $\alpha_k(\mathbf{x}, t)$ as follows:

$$\alpha(\mathbf{x}, t)\mathbf{u}_p(\mathbf{x}, t) = \sum_{k=1}^N \alpha_k(\mathbf{x}, t)[\mathbf{V}_k(t) + \boldsymbol{\Omega}_k(t) \times \{\mathbf{x} - \mathbf{X}_k(t)\}]. \quad (14)$$

The total (fluid + particle) velocity field $\mathbf{u}(\mathbf{x}, t)$ is then given by

$$\mathbf{u}(\mathbf{x}, t) = (1 - \alpha)\mathbf{u}_f + \alpha\mathbf{u}_p, \quad (15)$$

where $\mathbf{u}_f$ is the fluid velocity field.

Using Eq. (3) without the body force term, followed by Eq. (6), the intermediate total velocity field $\mathbf{u}^*(\mathbf{x}, t + \Delta t)$ is predicted. To enforce Eq. (15) and the solid–fluid impermeability condition, the hydrodynamic body force $\alpha\mathbf{f}^H$ [in Eq. (3)] is computed as

$$\alpha\mathbf{f}^H = \alpha(\mathbf{x}, t + \Delta t) \rho \text{Sh} \frac{\mathbf{u}_p(\mathbf{x}, t + \Delta t) - \mathbf{u}^*(\mathbf{x}, t + \Delta t)}{\Delta t}, \quad (16)$$

where ρ is the fluid density. Although $\rho = 1$ in the incompressible model, it is explicitly retained to avoid ambiguity, except in Sec. II A.

The hydrodynamic force $\mathbf{F}_k^H$ and torque $\mathbf{T}_k^H$ acting on the k -th particle are calculated based on the momentum exchange between the particle and suspending fluid over a time interval Δt [25]:

$$\mathbf{F}_k^H(t) = \mathbf{F}_k^{H,\text{tot}}(t) + \mathbf{F}_k^{H,\text{in}}(t), \quad (17)$$

$$\mathbf{T}_k^H(t) = \mathbf{T}_k^{H,\text{tot}}(t) + \mathbf{T}_k^{H,\text{in}}(t). \quad (18)$$

The terms $\mathbf{F}_k^{H,\text{tot}}$ and $\mathbf{T}_k^{H,\text{tot}}$ are obtained using the volume integration of the hydrodynamic body force contributing to the k -th particle, $\alpha_k\mathbf{f}^H$, over the computational domain:

$$\mathbf{F}_k^{H,\text{tot}}(t) = - \int \alpha_k\mathbf{f}^H(\mathbf{x}, t)dV, \quad (19)$$

$$\mathbf{T}_k^{H,\text{tot}}(t) = - \int [\mathbf{x} - \mathbf{X}_k(t)] \times \alpha_k\mathbf{f}^H(\mathbf{x}, t)dV. \quad (20)$$

The terms $\mathbf{F}_k^{H,\text{in}}$ and $\mathbf{T}_k^{H,\text{in}}$ compensate for the force and torque numerically induced by enforcing the artificial fluid within the particle region to match the particle velocity field. These are calculated from the time derivatives of the linear and rotational momenta of the internal artificial fluid as

$$\mathbf{F}_k^{H,\text{in}}(t) = \frac{\rho}{\rho_p} M_p \text{Sh} \frac{d\mathbf{V}_k(t)}{dt}, \quad (21)$$

$$\mathbf{T}_k^{H,\text{in}}(t) = \frac{\rho}{\rho_p} I_p \text{Sh} \frac{d\boldsymbol{\Omega}_k(t)}{dt}, \quad (22)$$

where ρ_p is the particle density. By eliminating the effects of the internal fluid mass, the iSP method enables the more precise calculation of the hydrodynamic forces and torques, thereby achieving stable and accurate simulations of particle dynamics over a wide range of flow conditions [25–27].

In particle-resolved simulations, lubrication forces between closely spaced particles are naturally included in the hydrodynamic forces exerted by the surrounding fluid. However, when the interparticle distance becomes smaller than approximately one lattice spacing, these forces tend to be underestimated. One approach to address this limitation is to explicitly compute the lubrication forces as functions of the interparticle distance and relative velocity, and incorporate them into the particle motion equations [Eqs. (9) and (10)]. Despite this, in several studies using particle-resolved simulations, researchers have reported good agreement with experimental data, even without such corrections [15, 17]. Therefore, no universally accepted guideline exists regarding the treatment of lubrication forces. In this study, the lubrication forces are not explicitly considered to keep the method simple.

D. Lees–Edwards boundary conditions for the 3D iSP-LBM-DEM method

To impose a shear flow in the x -direction without physical walls, the LE boundary conditions [28] are implemented in the 3D iSP-LBM-DEM method. Periodic replicas of the computational domain are placed above and below the primary domain, whereas standard periodic boundary conditions are applied in the x -direction. The upper and lower replica domains are translated at constant velocities $+U_{LE}$ and $-U_{LE}$ in the x -direction, respectively, which induces a shear flow with a shear rate of $\dot{\gamma} = U_{LE}/L_z$. Under this boundary condition, fluid and suspended particles exiting the computational domain through one LE boundary re-enter through the opposite LE boundary, with their positions and velocities shifted by $\pm U_{LE}t/Sh$ and $\pm U_{LE}$ in the x -direction, respectively. For clarity, in the following description, periodic treatments in the lateral (x - and y -) directions are omitted.

1. Fluid dynamics

Consider virtual fluid particles that exit through the lower LE boundary and re-enter from the upper LE boundary. After the collision step in the LBM, the post-collision distribution functions $f'_i(x, y, \Delta x)$ for $i = 7, 9, 10, 13, 14$ at $z = \Delta x$ are mapped to $z = L_z + \Delta x$ with a spatial shift of $+U_{LE}t/Sh$ and a velocity shift of $+U_{LE}$ in the x -direction [29]:

$$\tilde{f}'_i\left(x + \frac{U_{LE}t}{Sh}, y, L_z + \Delta x\right) = f'_{i,+U_{LE}}(x, y, \Delta x), \quad (23a)$$

$$f'_{i,+U_{LE}}(x, y, \Delta x) = f'_i(x, y, \Delta x) + f_i^{eq}(\mathbf{u}(x, y, \Delta x) + U_{LE}\mathbf{e}_x) - f_i^{eq}(\mathbf{u}(x, y, \Delta x)), \quad (23b)$$

where $\mathbf{e}_x$ is the unit vector in the x -direction.

Because the shift distance $U_{LE}t/Sh$ is generally not an integer multiple of the lattice spacing Δx , the mapped distribution functions typically lie between lattice nodes. To obtain the distribution functions at the lattice nodes that align with the primary domain, interpolation is required using the shifted distribution functions $\tilde{f}'_i(x + U_{LE}t/Sh, y, L_z + \Delta x)$. The shift distance $U_{LE}t/Sh$ ($\equiv D_{LE}$) is decomposed into an integer part D_{LE}^{int} and a fractional part D_{LE}^{frac} , where $D_{LE}^{int}/\Delta x$ is an integer and $D_{LE}^{frac}/\Delta x \in [0, 1)$. In most previous studies [29–31], researchers used first-order linear interpolation to obtain the distribution functions at the aligned lattice nodes:

$$\begin{aligned} f'_i(x + D_{LE}^{int}, y, L_z + \Delta x) &= \frac{D_{LE}^{frac}}{\Delta x} \tilde{f}'_i(x + D_{LE} - \Delta x, y, L_z + \Delta x) + \left(1 - \frac{D_{LE}^{frac}}{\Delta x}\right) \tilde{f}'_i(x + D_{LE}, y, L_z + \Delta x) \\ &= \frac{D_{LE}^{frac}}{\Delta x} f'_{i,+U_{LE}}(x - \Delta x, y, \Delta x) + \left(1 - \frac{D_{LE}^{frac}}{\Delta x}\right) f'_{i,+U_{LE}}(x, y, \Delta x). \end{aligned} \quad (24)$$

In this study, a third-order interpolation is used, which improves numerical accuracy as demonstrated in Sec. III A. The distribution functions at the aligned lattice nodes are given by

$$\begin{aligned} f'_i(x + D_{LE}^{int}, y, L_z + \Delta x) &= -\frac{1}{6} \frac{D_{LE}^{frac}}{\Delta x} \left(1 + \frac{D_{LE}^{frac}}{\Delta x}\right) \left(1 - \frac{D_{LE}^{frac}}{\Delta x}\right) f'_{i,+U_{LE}}(x - 2\Delta x, y, \Delta x) \\ &\quad + \frac{1}{2} \frac{D_{LE}^{frac}}{\Delta x} \left(1 + \frac{D_{LE}^{frac}}{\Delta x}\right) \left(2 - \frac{D_{LE}^{frac}}{\Delta x}\right) f'_{i,+U_{LE}}(x - \Delta x, y, \Delta x) \\ &\quad + \frac{1}{2} \left(1 + \frac{D_{LE}^{frac}}{\Delta x}\right) \left(1 - \frac{D_{LE}^{frac}}{\Delta x}\right) \left(2 - \frac{D_{LE}^{frac}}{\Delta x}\right) f'_{i,+U_{LE}}(x, y, \Delta x) \\ &\quad - \frac{1}{6} \frac{D_{LE}^{frac}}{\Delta x} \left(1 - \frac{D_{LE}^{frac}}{\Delta x}\right) \left(2 - \frac{D_{LE}^{frac}}{\Delta x}\right) f'_{i,+U_{LE}}(x + \Delta x, y, \Delta x). \end{aligned} \quad (25)$$

Similarly, for virtual fluid particles crossing from the upper to the lower LE boundary, the post-collision distribution functions $f'_i(x, y, L_z)$ for $i = 6, 8, 11, 12, 15$ at $z = L_z$ are mapped to $z = 0$ with a spatial shift of $-U_{LE}t/Sh$ and velocity shift of $-U_{LE}$ in the x -direction:

$$\tilde{f}'_i\left(x - \frac{U_{LE}t}{Sh}, y, 0\right) = f'_{i,-U_{LE}}(x, y, L_z), \quad (26a)$$

$$f'_{i,-U_{LE}}(x, y, L_z) = f'_i(x, y, L_z) + f_i^{\text{eq}}(\mathbf{u}(x, y, L_z) - U_{LE}\mathbf{e}_x) - f_i^{\text{eq}}(\mathbf{u}(x, y, L_z)). \quad (26b)$$

The distribution functions at the lattice nodes that align with the primary domain are obtained using the first-order linear interpolation:

$$f'_i(x - D_{LE}^{\text{int}}, y, 0) = \frac{D_{LE}^{\text{frac}}}{\Delta x} f'_{i,-U_{LE}}(x + \Delta x, y, L_z) + \left(1 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) f'_{i,-U_{LE}}(x, y, L_z), \quad (27)$$

or the third-order interpolation:

$$\begin{aligned} f'_i(x - D_{LE}^{\text{int}}, y, 0) = & -\frac{1}{6} \frac{D_{LE}^{\text{frac}}}{\Delta x} \left(1 + \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) \left(1 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) f'_{i,-U_{LE}}(x + 2\Delta x, y, L_z) \\ & + \frac{1}{2} \frac{D_{LE}^{\text{frac}}}{\Delta x} \left(1 + \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) \left(2 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) f'_{i,-U_{LE}}(x + \Delta x, y, L_z) \\ & + \frac{1}{2} \left(1 + \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) \left(1 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) \left(2 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) f'_{i,-U_{LE}}(x, y, L_z) \\ & - \frac{1}{6} \frac{D_{LE}^{\text{frac}}}{\Delta x} \left(1 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) \left(2 - \frac{D_{LE}^{\text{frac}}}{\Delta x}\right) f'_{i,-U_{LE}}(x - \Delta x, y, L_z). \end{aligned} \quad (28)$$

Note that in the actual computational implementation, Eqs. (23)–(28) are evaluated while accounting for the periodic boundaries in the flow direction, which are applied at $x = 0.5\Delta x$ and $L_x + 0.5\Delta x$. During the subsequent streaming step, the interpolated distribution functions $f'_i(x, y, 0)$ and $f'_i(x, y, L_z + \Delta x)$ are streamed into the primary computational domain accordingly.

2. Particle dynamics

When the center of mass of the k -th particle crosses the lower or upper LE boundary, it re-enters from the opposite side with a spatial shift of $\pm U_{LE}t/Sh$ and velocity shift of $\pm U_{LE}$ in the x -direction:

$$X_k = X_k \pm \frac{U_{LE}t}{Sh}, \quad (29)$$

$$V_{kx} = V_{kx} \pm U_{LE}, \quad (30)$$

where X_k and V_{kx} denote the x -components of the particle position $\mathbf{X}_k$ and velocity $\mathbf{V}_k$, respectively.

When the contact force and torque exerted on the k -th particle by the l -th particle are computed, an image particle located in the upper or lower replica domain is also taken into account in addition to the original (l -th) particle. This is similar to the periodic boundary condition; however, the x -positions and x -velocities of the image particles are shifted in the same manner as in Eqs. (29) and (30), respectively.

3. Fluid–particle coupling

When a part of the k -th particle crosses the lower or upper LE boundary, it reappears on the opposite LE boundary. This is represented by the introduction of a corresponding image particle, denoted by the k^+ -th or k^- -th particle, located in the upper or lower replica domain, respectively. The x -position and x -velocity of the image particle are shifted in the same manner as in Eqs. (29) and (30), respectively.

The particle field [Eq. (13)] and particle velocity field [Eq. (14)] are then computed by considering both the original particles and, when necessary, their image particles:

$$\alpha(\mathbf{x}, t) = \sum_{k=1}^N (\alpha_k(\mathbf{x}, t) + \alpha_{k^\pm}(\mathbf{x}, t)), \quad (31)$$

$$\begin{aligned} \alpha(\mathbf{x}, t) \mathbf{u}_p(\mathbf{x}, t) = & \sum_{k=1}^N \{ \alpha_k(\mathbf{x}, t) [\mathbf{V}_k(t) + \boldsymbol{\Omega}_k(t) \times (\mathbf{x} - \mathbf{X}_k(t))] \\ & + \alpha_{k^\pm}(\mathbf{x}, t) [\mathbf{V}_{k^\pm}(t) + \boldsymbol{\Omega}_k(t) \times (\mathbf{x} - \mathbf{X}_{k^\pm}(t))] \}, \end{aligned} \quad (32)$$

where $\alpha_{k^\pm}(\mathbf{x}, t)$ denotes the SP function at position $\mathbf{x}$ and time t corresponding to the image particle, and $\mathbf{X}_{k^\pm}(t)$ and $\mathbf{V}_{k^\pm}(t)$ denote its position and velocity, respectively.

Accordingly, using $\alpha(\mathbf{x}, t)$ and $\alpha(\mathbf{x}, t) \mathbf{u}_p(\mathbf{x}, t)$, which include contributions from both the original and image particles, the hydrodynamic body force $\alpha \mathbf{f}^H$ is computed using Eq. (16). The hydrodynamic force $\mathbf{F}_k^H$ and torque $\mathbf{T}_k^H$ acting on the k -th particle are also calculated based on Eq. (17) together with Eqs. (19) and (21), and Eq. (18) together with Eqs. (20) and (22), respectively. However, it should be noted that the terms $\mathbf{F}_k^{H, \text{tot}}$ and $\mathbf{T}_k^{H, \text{tot}}$ must be volume-integrated over the regions corresponding to both the original and its image particles within the computational domain.

4. Evaluation of the apparent viscosity of the suspension

When a shear flow is imposed using the LE boundary conditions, the apparent viscosity of the suspension, η_s , is evaluated from the bulk shear stress σ as

$$\eta_s = \frac{\sigma}{\dot{\gamma}}. \quad (33)$$

In the present system, the bulk shear stress σ includes contributions from the viscous behavior of suspending fluid, hydrodynamic interactions between the fluid and particles, and physical contacts between particles [36–38], and is expressed as

$$\begin{aligned} \sigma &= \eta \dot{\gamma} + \sigma^H + \sigma^C \\ &= \eta \dot{\gamma} - \frac{1}{V} \sum_k \int_{\Omega_k^\xi} [\alpha_k \mathbf{f}^H(x, y, z)]_x (z - Z_k) dV + \frac{1}{V} \sum_k \sum_{\delta_{kl} \geq 0} [\mathbf{F}_{kl}^C]_x (Z_l - Z_k), \end{aligned} \quad (34)$$

where η is the suspending fluid viscosity, Ω_k^ξ denotes the interfacial region of the k -th particle, $[\cdot]_x$ denotes the x -component of a vector, and Z_k and Z_l are the z -coordinates of the centers of the k -th and l -th particles, respectively.

Accordingly, the total apparent viscosity, η_s , can be decomposed into the three components arising from the suspending fluid viscosity, fluid–particle hydrodynamic interactions, and particle–particle contacts:

$$\eta_s = \frac{\eta \dot{\gamma} + \sigma^H + \sigma^C}{\dot{\gamma}} = \eta + \eta^H + \eta^C. \quad (35)$$

E. Computational conditions and parameters

The computational conditions and parameters used throughout the simulations are summarized below. In the LBM, the relaxation time in Eq. (3) was set to $\tau = 0.86$, corresponding to a kinematic viscosity of the suspending fluid of $\nu = \eta/\rho = 0.12\Delta x$. The thickness of the SP function was $\xi = 2\Delta x$, which provides sufficient accuracy in terms of the hydrodynamic forces acting on particles [25].

In the DEM, the normal and tangential spring constants were $k_n = 10\,000\Delta x$ and $k_t = 0.25k_n$, respectively, which ensured particle rigidity. The corresponding damping coefficients were $d_n = 2\sqrt{M_p k_n}$ and $d_t = 0.5d_n$. The interparticle friction coefficient μ_c was varied between 0 and 1. The

particle surface roughness height was $h_r/a = 5 \times 10^{-3}$. Because the stable time step required for the DEM is typically smaller than Δt , the DEM time step (Δt_{DEM}) was set to $\Delta t/50$ based on preliminary stability tests. Under these conditions, the maximum particle overlap observed in all simulations remained smaller than the surface roughness height h_r .

III. RESULTS

The following sections first verify key aspects of the proposed method, including the implementation of the LE boundary conditions (Sec. III A), where appropriate numerical parameters are determined, and the method for evaluating suspension viscosity (Sec. III B). In addition to these fundamental verification tests, Sec. III C applies this method to simulations of suspension shear flows over a wide range of particle volume fractions and interparticle friction coefficients, and compares the results with previous experimental and numerical studies. Overall, these results confirm the validity, robustness, and applicability of the proposed method to a broad range of suspension conditions.

A. Verification of the LE boundary conditions implementation

To verify the implementation of the LE boundary conditions, the motion of both single and multiple particles crossing the LE boundary was simulated. Similar benchmark tests were conducted in previous studies using 2D simulations of particles denser than the suspending fluid [30–32]. In the present study, we first conduct the analysis in a 3D system with a single dense particle, and then further extend it to particles with the same density as the suspending fluid.

The computational conditions in this section were based on those in Ref. [30]. The computational domain was set to $L_x \times L_y \times L_z = 128\Delta x \times 100\Delta x \times 128\Delta x$. Periodic boundary conditions were applied in the x - and y -directions, whereas the LE boundary conditions were applied in the z -direction to generate shear flow along the x -direction, superimposed on a constant background velocity $\mathbf{u}_b = (0.025, 0, 0.005)$. The flow field was initialized and maintained at a uniform shear rate of $\dot{\gamma} = U_{\text{LE}}/L_z = 1.25 \times 10^{-5}(\Delta x)^{-1}$. Particles were initially placed along a line parallel to the x axis at the center of the computational domain, with their translational velocities set equal to the local fluid velocity, that is, $\mathbf{V}_k(t=0) = \mathbf{u}_b$, and their angular velocities set to $\mathbf{\Omega}_k(t=0) = (0, 0.5\dot{\gamma}, 0)$. The particles were then allowed to move freely. As they were advected by the flow, they exited the computational domain through the upper LE boundary and re-entered through the lower LE boundary.

1. Single particle denser than the suspending fluid

First, the performance of the first- and third-order interpolation schemes was examined using the simulation of the motion of a single particle with a radius of $a = 6\Delta x$ and density of $\rho_p = 10$. In the single-particle cases, unless otherwise stated, the particle was placed at the center of the computational domain. Figure 3(a) compares the time evolution of the particle's vertical velocity V_{1z} obtained using the two schemes. The times at which the particle's center of mass crossed the LE boundary are indicated by vertical dotted lines at $\dot{\gamma}t/\text{Sh} = 0.16$ and 0.48 . For both schemes, the particle moved vertically at a constant velocity equal to $V_{1z}(t=0) = 0.005$, and the vertical velocity remained unchanged before and after crossing the LE boundary.

In previous 2D simulations, the angular velocity of a particle was shown to be particularly sensitive to the accuracy of the method [32]. Therefore, the time evolution of the particle's angular velocity about the y axis, Ω_{1y} , obtained using the two interpolation schemes, is plotted in Fig. 3(b). For both schemes, Ω_{1y} initially increased slightly to approximately $1.001 \times 0.5\dot{\gamma}$ and then remained nearly constant. However, dips in Ω_{1y} were observed as the particle crossed the LE boundary. The magnitude of these dips for the third-order interpolation scheme was less than one-tenth of that for the first-order interpolation scheme. These results indicate that the third-order interpolation scheme

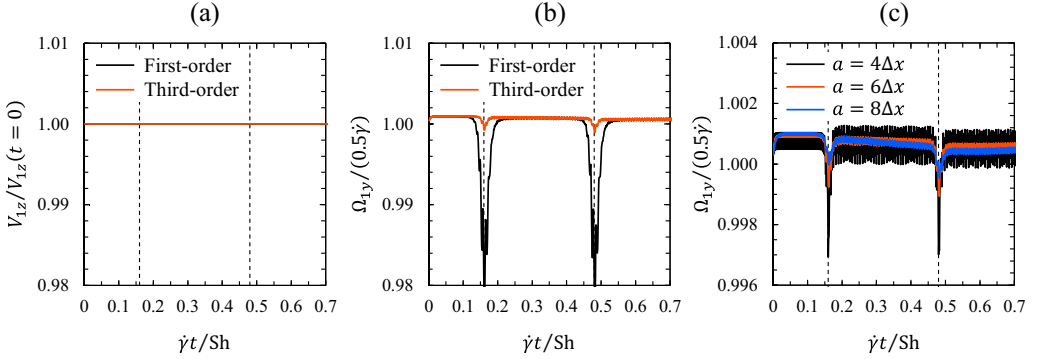


FIG. 3. Motion of a single particle crossing the LE boundary with a particle density of $\rho_p = 10$. Time evolutions of (a) the vertical velocity V_{1z} and (b) the angular velocity about the y axis, Ω_{1y} , for a particle with a radius of $a = 6\Delta x$, computed using first- and third-order interpolation schemes. (c) Time evolutions of Ω_{1y} for a particle with different radii, $a = 4\Delta x$, $6\Delta x$, and $8\Delta x$, computed using the third-order interpolation scheme.

was effective in maintaining numerical accuracy in 3D simulations; therefore, it was adopted in the subsequent calculations.

Next, the effect of particle resolution was examined using the third-order interpolation scheme. Figure 3(c) shows the time evolutions of Ω_{1y} for different particle sizes, $a = 4\Delta x$, $6\Delta x$, and $8\Delta x$, with a particle density of $\rho_p = 10$. The y axis in this plot has been rescaled to better highlight the deviations between particle resolutions. For all three cases, dips in Ω_{1y} were observed as the particle crossed the LE boundary. Their magnitudes were smaller for larger particle resolutions, specifically with that for $a = 6\Delta x$ being less than 0.2%. The rescaled plot also reveals small oscillatory behaviors in Ω_{1y} , which originated from numerical perturbations associated with the fluid–particle coupling. These oscillations appeared from the onset of particle motion and were slightly amplified as the particle crossed the LE boundary, where interpolation of the distribution functions was required during the fluid–particle coupling procedure. Nevertheless, the amplitude of these oscillations remained below 0.02% for the case of $a = 6\Delta x$. From these results, the particle size $a = 6\Delta x$ was adopted in this study, considering the balance between sufficient accuracy and computational efficiency.

2. Single and multiple particles with the same density as the suspending fluid

Systems in which the particles and the suspending fluid have identical densities are generally challenging to simulate stably. When the LE boundary conditions are used, numerical errors inevitably arise as the particles cross the boundary, which can further compromise the stability of the simulation. Consequently, the motion of particles with the same density as the suspending fluid ($\rho_p = \rho = 1$) was simulated to further confirm the robustness of our method.

First, the motion of a single particle was analyzed. Figure 4 compares the time evolutions of the angular velocity Ω_{1y} for a single particle with different densities $\rho_p = 1$ and $\rho_p = 10$. The particle motions in both cases were very similar, showing nearly identical magnitudes of dips in Ω_{1y} , except for the oscillatory behavior. For $\rho_p = 1$, the oscillation was amplified after the particle crossed the LE boundary, with an amplitude approximately 5 times that observed for $\rho_p = 10$. This is expected because a lighter particle responds more strongly to the perturbations associated with the fluid–particle coupling procedure at the LE boundaries. Even for this lighter particle, the magnitude of the dips remained below 0.3%, and the oscillation amplitude remained below 0.1%, both of which are sufficiently small to be considered negligible.

Next, multiparticle cases with $N = 2$ and 4 particles were considered. When these particles were initially positioned along a line parallel to the x axis at the center of the computational domain

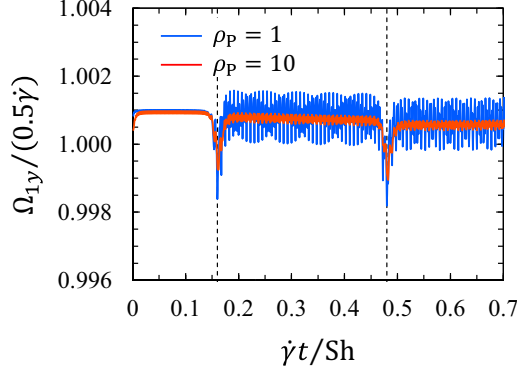


FIG. 4. Time evolutions of the angular velocity about the y axis, Ω_{1y} , for a single particle with a radius of $a = 6\Delta x$ under the LE boundary conditions, for two different densities: $\rho_p = 1$ and $\rho_p = 10$.

with equal spacing, their time evolution was expected to be synchronous in theory [31]. In each N -particle simulation, the initial x -position of the k -th particle was given by $X_k(t = 0) = L_x(k - 0.5)/N + 0.5\Delta x$.

Figure 5 shows the time evolutions of the angular velocities Ω_{ky} of the k -th particle for the two- and four-particle cases, including the single-particle result for comparison. In both multiparticle cases, the angular velocities decreased from the initial value of $0.5\dot{\gamma}$ and then remained nearly constant. The equilibrium values of Ω_{ky} decreased with increasing particle number N , which is expected because the particles interact more strongly with neighbors at smaller interparticle distances. These behaviors are consistent with 2D simulations of particles denser than the suspending fluid [31]. In each multiparticle case, the angular velocities of all particles were in exact agreement within numerical uncertainty, indicating synchronous motion as predicted theoretically. Furthermore, the magnitudes of dips and the oscillation amplitudes were nearly identical to those observed in the single-particle simulation.

As shown here, the motion of single and multiple particles with the same density as the suspending fluid crossing the LE boundary was successfully simulated. These results confirm that the LE boundary conditions are correctly implemented in our simulation method, maintaining stability and

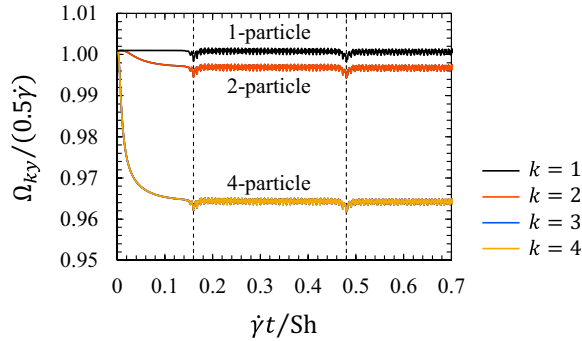


FIG. 5. Time evolutions of the angular velocities about the y axis, Ω_{ky} , of the k -th particle for the single-, two-, and four-particle systems with a radius of $a = 6\Delta x$ and a density of $\rho_p = 1$ under the LE boundary conditions. For each multiparticle case, the data for all particles are plotted, and the values agree exactly within numerical uncertainty.

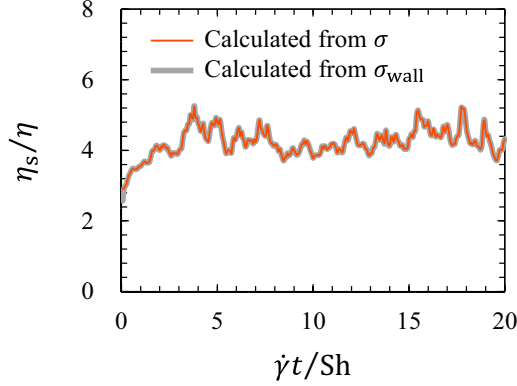


FIG. 6. Comparison of two relative apparent viscosities, η_s/η , evaluated from the bulk shear stress, σ , and shear stress acting on the moving walls, σ_{wall} , for a suspension with a particle volume fraction of $\phi = 0.35$ and an interparticle friction coefficient of $\mu_c = 0.3$ in a shear flow field driven by oppositely moving walls. Each plotted value represents a time average over a period of $2000\Delta t$, corresponding to $\Delta(\dot{\gamma}t/\text{Sh}) \approx 6.667 \times 10^{-2}$.

accuracy. In what follows, all simulations were conducted using particles with the same density as the suspending fluid to further demonstrate the robustness of our method.

B. Verification of the method for evaluating suspension viscosity

Because the method for evaluating the apparent viscosity of suspensions from the bulk shear stress, which is averaged over the entire computational domain using Eq. (34), is nontrivial, its verification is essential. To this end, physical walls were placed at the top and bottom of the simulation domain and moved in opposite directions at a constant speed U_w to generate a shear flow [39], instead of applying the LE boundary conditions. In this setup, the apparent viscosity of the suspension can be evaluated from both the bulk shear stress, σ , and the shear stress acting on the moving walls, σ_{wall} [39]. Although the instantaneous values of these two viscosities may differ at a given time, their time-averaged values are expected to be essentially equal.

The simulation setup used here was the same as that in the main simulations described later in Sec. III C (see Fig. 1), except for the moving-wall conditions in the z -direction boundaries. The wall speed was set to $U_w = U_{\text{LE}}/2 = 0.001$, which resulted in a shear rate of $\dot{\gamma} = 2U_w/L_z \approx 3.333 \times 10^{-5}(\Delta x)^{-1}$. The particle Reynolds number of the system was then $\text{Re}_p = \rho\dot{\gamma}a^2/\eta = 0.01$.

The apparent viscosity, η_s/η , evaluated from both the bulk shear stress, σ , and the shear stress acting on the moving walls, σ_{wall} , was averaged over time intervals of $2000\Delta t$, corresponding to $\Delta(\dot{\gamma}t/\text{Sh}) \approx 6.667 \times 10^{-2}$. As an example, the two relative apparent viscosities η_s/η for a suspension with a particle volume fraction of $\phi = 0.35$ and an interparticle friction coefficient of $\mu_c = 0.3$ are shown in Fig. 6. As shown in the figure, the time-averaged relative viscosities obtained from σ and σ_{wall} were in excellent agreement. This result confirms that the method for evaluating the suspension's apparent viscosity from the bulk shear stress is correctly implemented in the proposed method.

C. Simulations of suspension shear flow over a wide range of particle volume fractions

1. Comparison with previous experimental and numerical results

The dependence of the suspension's apparent viscosity on the particle volume fraction ϕ was investigated by simulating shear flow in the system shown in Fig. 1. The dimensions of the simulation domain were set to $L_x \times L_y \times L_z = 20a \times 10a \times 10a$, where the particle radius was $a = 6\Delta x$, as determined in Sec. III A. Periodic boundary conditions were applied in the x - and

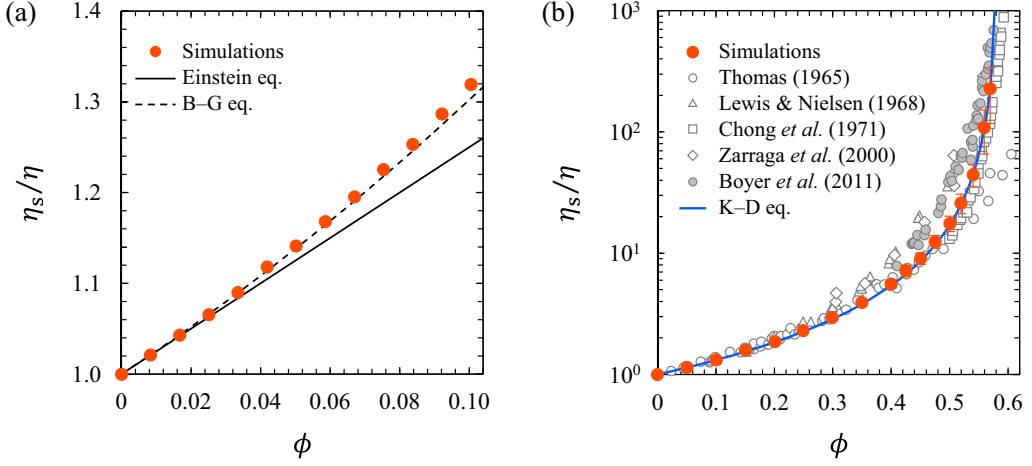


FIG. 7. Relative viscosities η_s/η of particle suspensions as a function of particle volume fraction ϕ , obtained from shear flow simulations with an interparticle friction coefficient of $\mu_c = 0.3$. (a) Data for $\phi < 0.104$. The solid and dotted lines represent theoretical predictions given by the Einstein equation ($\eta_s/\eta = 1 + 2.5\phi$) [40] and Batchelor–Green equation ($\eta_s/\eta = 1 + 2.5\phi + 5.2\phi^2$) [41], respectively. Error bars are omitted because they are too small to be clearly visible. (b) Data over a wide range of ϕ . Gray markers indicate experimental data from previous studies [42–46], whereas the solid line corresponds to the semiempirical prediction given by the Krieger–Dougherty equation [Eq. (36)] [47] using $\phi_m = 0.583$, which was determined by fitting the equation to the simulation data.

y-directions, whereas the LE boundary conditions were applied in the z-direction to generate shear flow along the x-direction. The flow field was initialized and maintained at a uniform shear rate of $\dot{\gamma} = U_{LE}/L_z$. In the computational domain, the particles were randomly placed such that they were not in contact with each other, with their translational velocity set to a value equal to the respective local fluid velocity and their angular velocity set to $\mathbf{\Omega}_k(t=0) = (0, 0.5\dot{\gamma}, 0)$, and then allowed to move freely. The particle density was set to $\rho_p = \rho = 1$. The imposed shear rate was $\dot{\gamma} \approx 3.333 \times 10^{-5}(\Delta x)^{-1}$, corresponding to a particle Reynolds number of $Re_p = \rho\dot{\gamma}a^2/\eta = 0.01$.

The time evolution of the apparent viscosity was computed by averaging over time intervals of $2000\Delta t$, which corresponded to $\Delta(\dot{\gamma}t/Sh) \approx 6.667 \times 10^{-2}$. In all the cases examined, the flow was considered to have reached a steady state for $\dot{\gamma}t/Sh \geq 10$, although significant fluctuations in the apparent viscosity were observed at high volume fractions. Accordingly, the steady-state apparent viscosities reported below were obtained by averaging over the interval $10 \leq \dot{\gamma}t/Sh \leq 20$. Error bars indicate the corresponding standard deviations.

Suspension flows were examined at low volume fractions with the interparticle friction coefficient fixed at $\mu_c = 0.3$. Under these conditions, the initial particle configuration significantly influenced the apparent viscosity because of the limited number of particles. To mitigate this effect, particles were randomly positioned such that the center-to-center distance between any two particles exceeded $0.8(\frac{L_x L_y L_z}{N_p})^{\frac{1}{3}}$. Figure 7(a) shows the relative viscosities η_s/η of suspensions obtained from shear flow simulations at particle volume fractions $\phi < 0.104$ as a function of ϕ . For comparison, theoretical predictions based on the Einstein equation, $\eta_s/\eta = 1 + 2.5\phi$ [40], and the Batchelor–Green equation, $\eta_s/\eta = 1 + 2.5\phi + 5.2\phi^2$ [41], are also plotted. The simulation results demonstrated good agreement with the Einstein equation for $\phi < 0.02$ and with the Batchelor–Green equation for $\phi < 0.06$, which indicates that the proposed method accurately captured the shear flow behavior of suspensions in dilute regimes under the LE boundary conditions.

Suspension flows were examined over a wide range of volume fractions, keeping the interparticle friction coefficient fixed at $\mu_c = 0.3$. Figure 7(b) shows the relative viscosities, η_s/η , of suspensions

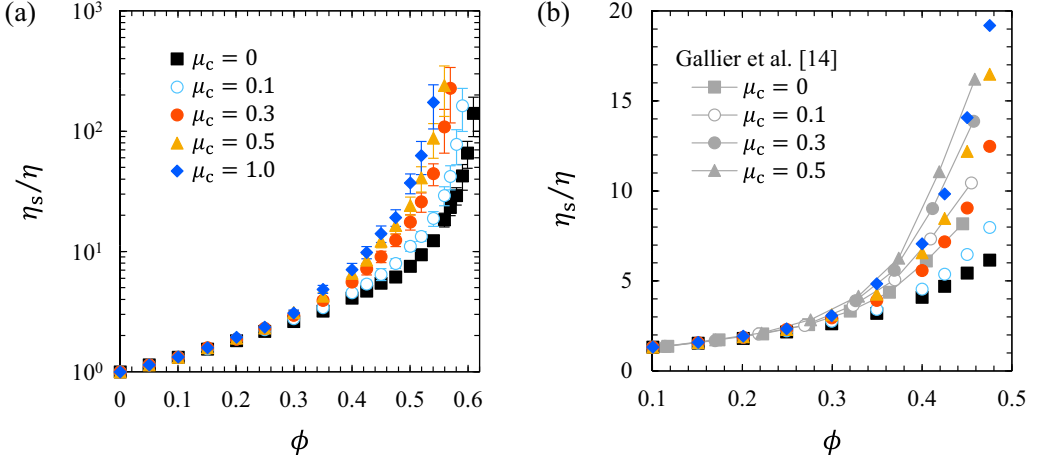


FIG. 8. (a) Relative viscosity η_s/η of particle suspensions as a function of particle volume fraction ϕ for shear flow simulations conducted with various friction coefficients μ_c . (b) Comparison between the results obtained by the proposed method and the numerical data reported by Gallier *et al.* [14] (gray markers), who used a fictitious domain approach for particle-resolved simulations.

as a function of ϕ , with the vertical axis plotted on a logarithmic scale. For comparison, typical experimental data from previous studies using suspensions of noncolloidal spherical particles dispersed in a Newtonian fluid [42–46] are also shown. The logarithmic scale reduces the apparent differences between the studies, but these experimental data exhibited variability in relative viscosity at nearly identical volume fractions. This variability was small at low volume fractions but increased significantly at higher volume fractions. One possible contributing factor is the interparticle friction coefficients, whose exact values are generally difficult to measure, may have differed among these experimental studies. In addition, differences in measurement methods and experimental setups may also have contributed to this variability. The simulation results demonstrated good agreement with all experimental data for $\phi < 0.3$ and matched the lower bound of the experimental range for $0.3 < \phi < 0.5$. In the highly concentrated regime ($\phi > 0.5$), the simulation results agreed well with the data reported by Chong *et al.* [44].

The simulation results were also compared with the semiempirical prediction given by the Krieger–Dougherty equation [47]:

$$\frac{\eta_s}{\eta} = \left(1 - \frac{\phi}{\phi_m}\right)^{-[\eta]\phi_m}, \quad (36)$$

where ϕ_m is the maximum packing fraction at which particles jam and flow ceases, and $[\eta]$ is the intrinsic viscosity, typically taken as 2.5 for spherical particles. The Krieger–Dougherty equation provides a good description of the relationship between relative viscosity and particle volume fraction over a wide range, provided that an appropriate value of ϕ_m is used. Although several studies have explored methods to estimate ϕ_m from the suspension characteristics, no universally accepted procedure currently exists. Accordingly, in the present study, ϕ_m was determined by fitting the Krieger–Dougherty equation to the simulation data. Using the resulting value of $\phi_m = 0.583$, the simulation results showed good agreement with the prediction of Eq. (36).

The good agreement of the numerical results with both the experimental data and the prediction by the Krieger–Dougherty equation demonstrated that the proposed method successfully captured the shear flow behavior of suspensions over a wide range of volume fractions.

Next, the effect of the interparticle friction coefficient, μ_c , on the apparent viscosity of the suspension, η_s , was examined. Figure 8(a) shows the relative viscosity, η_s/η , as a function of the

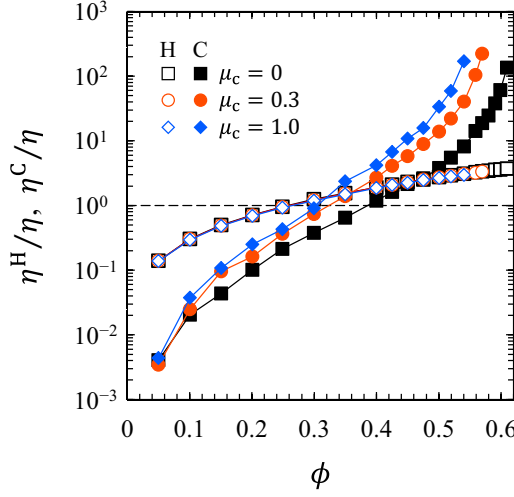


FIG. 9. Viscosity components arising from fluid–particle hydrodynamic interactions and particle–particle contacts, normalized by the suspending fluid viscosity as η^H/η and η^C/η , respectively, as functions of particle volume fraction ϕ for different interparticle friction coefficients $\mu_c = 0, 0.3$, and 1.0 .

particle volume fraction ϕ for various values of μ_c . For $\phi < 0.3$, the effect of μ_c on the relative viscosity was negligible. However, at higher ϕ , the relative viscosity increased as μ_c increased. In particular, at $\phi \geq 0.55$, the relative viscosity varied by nearly an order of magnitude depending on μ_c .

Figure 8(b) compares the results obtained by the proposed method with the numerical data reported by Gallier *et al.* [14], who used a fictitious domain approach for particle-resolved simulations. At low volume fractions, both datasets demonstrated good agreement and little sensitivity to the interparticle friction coefficient. For $\phi > 0.3$, although the proposed method predicted slightly lower viscosities than those reported by Gallier *et al.*, the dependence on the friction coefficient was qualitatively similar. These findings indicate that the proposed method successfully captured the effect of the friction coefficient on the shear flow behavior of suspensions. It is worth noting that the use of the LE boundary conditions enabled a significantly smaller computational domain to be used in the present study than that used in Ref. [14].

2. Insights provided by the proposed method

Based on Eq. (35), the total apparent viscosity, η_s , can be decomposed into three components, corresponding to the suspending fluid viscosity (η), the contribution from the fluid–particle hydrodynamic interactions (η^H), and the contribution from the particle–particle contacts (η^C). This allows us to explore the factors contributing to the variation in suspension viscosity [Figs. 7(b) and 8(a)].

Figure 9 shows the normalized contributions, η^H/η and η^C/η , as functions of the particle volume fraction ϕ for different interparticle friction coefficients, $\mu_c = 0, 0.3$, and 1.0 . Both the hydrodynamic and contact-related viscosities increased monotonically with increasing ϕ , but their rates of increase and dependence on μ_c differed significantly. At very low particle volume fractions, both the hydrodynamic contribution, η^H , and contact-related contribution, η^C , were smaller than the suspending fluid viscosity, η . As ϕ increased, η^H became comparable to η around $\phi = 0.25$. By contrast, η^C remained smaller than both η and η^H , as interparticle contacts were still rare because of the low particle number density. This observation is consistent with the earlier finding that μ_c had a negligible influence on the apparent viscosity for $\phi < 0.3$ [Fig. 8(a)]. As ϕ increased further, η^H increased relatively gradually, whereas η^C increased more sharply and eventually exceeded η^H . At even higher ϕ , η^C became significantly larger than both η and η^H . In this regime, η^C exhibited a

strong dependence on the interparticle friction coefficient μ_c : the larger the value of μ_c , the higher the value of η^C . By contrast, η^H varied with ϕ , but remained nearly independent of μ_c . These results indicate that the apparent viscosity of highly concentrated suspensions is predominantly governed by direct interparticle contacts.

As demonstrated in this study, the dominant factor that contributed to the suspension viscosity changed with the particle volume fraction, transitioning from the suspending fluid viscosity to the fluid–particle hydrodynamic interactions, and finally to the particle–particle direct contacts. This finding indicates that the particle-resolved approaches are particularly relevant in the moderately dense regime, where both hydrodynamic interactions and interparticle contacts significantly influence the suspension dynamics. More importantly, the proposed method is capable of simulating shear flows of suspensions over a wide range of volume fractions without modifying the method or tuning computational parameters. This allows the method to be applied to more realistic and industrially relevant systems, where spatial heterogeneity in particle concentration, arising from physicochemical interactions, is often observed.

IV. CONCLUSION

We presented a particle-resolved simulation method for analyzing the rheology of suspensions consisting of noncolloidal, monodisperse, spherical particles dispersed in an incompressible Newtonian fluid. The method employed the LBM for fluid dynamics and the DEM for particle dynamics, coupled through the iSP approach. To enable efficient simulations of suspensions under shear flow, the LE boundary conditions were implemented in the iSP-LBM-DEM.

Verification tests confirmed the correct implementations of the LE boundary conditions and the method for evaluating the apparent viscosity of suspensions, and determined appropriate numerical parameters. Applications to suspension shear flows over a wide range of particle volume fractions and interparticle friction coefficients showed good agreement with previous experimental and numerical results. The method provided a decomposition of the total apparent viscosity into its constituent components, demonstrating that the dominant factor influencing suspension viscosity changed with the particle volume fraction, from the suspending fluid viscosity to the fluid–particle hydrodynamic interactions, and finally to the particle–particle direct contacts. These findings demonstrate that the proposed method, including the reliable implementation of the LE boundary conditions, provides an efficient, robust, and accurate approach for numerical investigations of suspension rheology.

ACKNOWLEDGMENT

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible, and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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